

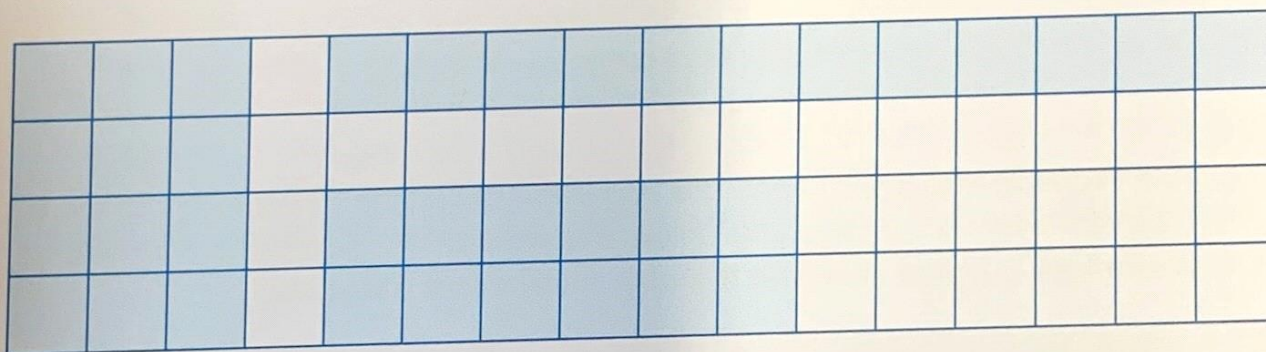
10. Multiplication with arrays

MULTIPLICATIVE THINKING

Multiplicative thinking underpins concepts such as area and volume and helps us to link fractions, decimals and percent. It is a fundamental idea that we will use to build other concepts.

Experiment

The diagrams below show rectangles or arrays formed from 12 squares drawn on grids. Use these diagrams to stimulate your own thinking and come up with as many different rectangles as you can formed from 36 squares. This is called the **Array Strategy** and it will be used frequently.



Explore

Each array that you have drawn represents a multiplication fact. Write the multiplication fact represented next to each array.

6x6 makes a square. Do you think this is still a rectangle? Discuss this with your group and check your thinking with your teacher. What arrays do you think you could make with 35 squares? How about with 37 squares?

Evaluate

Imagine that you turned each array by a quarter. What other multiplication facts are now represented? Do you think that this would work with all multiplication facts, or just these ones? Explain your thinking. Find the rectangle of 12 above that is 3x4.

Extend

The **Array Strategy** for multiplication can help us to multiply much larger numbers. Sketch an array to show 23×45 . How you could break your large rectangle into smaller parts to make multiplication easier? Draw the parts and show your thinking.

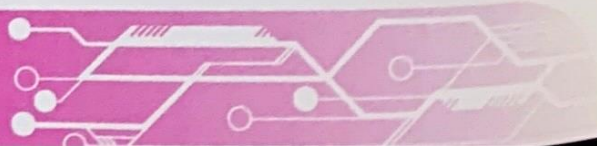
Both 3 and 4 are **factors** of 12. 12 is the **multiple** of 3 and 4. What factors can you find for 36? **Use An Array.**

Draw as many arrays as you can with 7 squares. Now try 11 squares. What do you notice? 7 and 11 are both **prime** numbers whereas 12 and 36 are **composite**. What do you think square numbers might look like?

Explain

Using the questions above to stimulate your thinking, describe the following terms in your own words. Include a diagram if it is useful.

- Factor
- Multiple
- Prime
- Composite



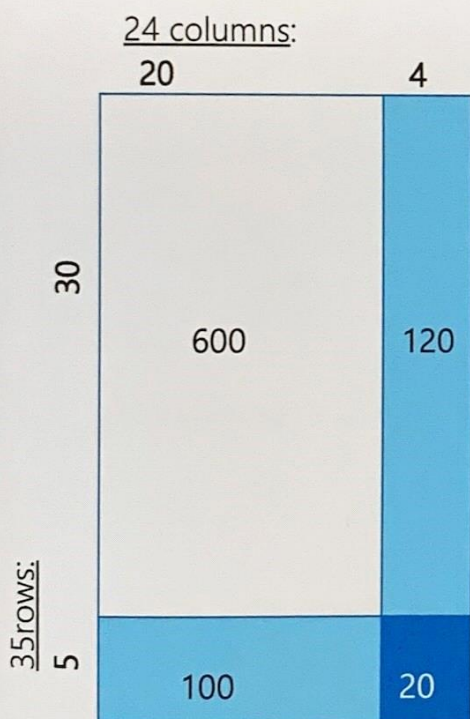
11. Efficient strategies for multiplication

 **MULTIPLICATIVE THINKING**  **PARTITIONING**

Many multiplication strategies that we were taught involve memorising a procedure then trying not to mess it up. This lesson is designed to teach more efficient strategies for multiplication, starting with the **Array Strategy**. This strategy is also great for approximation and easy to carry out mentally.

Strategy one: sketch an array and then partition

The following diagram shows an array of 24×35 . What do you think you might do with each part to work out the answer? The answer is 840. Use this to work out how to use the strategy.



Set A: Different numbers of digits

Try your strategy out on the following questions.

Your sketches do not need to be to scale.

1. 27×5
2. 45×34
3. 6×213

Set B: Whole numbers by decimal numbers

The same strategy of an array can help you to multiply decimal numbers easily. Remember if you get stuck to think of the decimal numbers as money.

4. 1.5×4
5. 2.2×6

Strategy two: partitioning arrays for difficult to remember facts

Sometimes, particular multiplication facts are easier to remember than others. Most students find 7s particularly difficult to memorise, and 7s do not have an easy counting pattern if you get stuck. Luckily, the strategy shown above for breaking arrays into parts helps with 7s.

Sketch an array of 7×6 . Now consider how you could partition the 7 to make easier parts. Draw it, show each part, and discuss how you cut the array with your friends. Did you do it the same way? What was easiest for you? How might you apply this same thinking to multiply a number by 12?

Strategy three: finger patterns for 6, 7, 8 and 9 facts

One final strategy is to use your fingers to quickly calculate 6s, 7s, 8s and 9s. This strategy was used in the middle ages for quick calculations. Your teacher will demonstrate the strategy. Discuss it with a partner and reflect on which of the strategies you think will be useful.

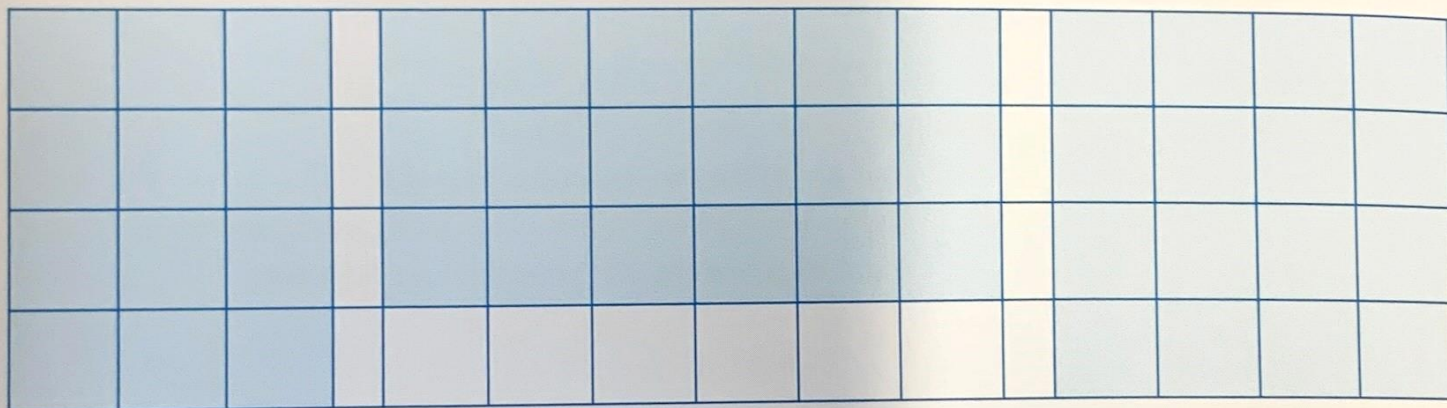
12. Application of arrays to area

MULTIPLICATIVE THINKING

Area is one of the most common calculations used in building, farming, renovating and town planning. You don't need a formula, but you do need to apply your understanding of arrays to work it out. Use the following diagram to deduce what area means and determine how to calculate the area of a rectangle. This lesson applies the **Array Strategy**.

Experiment

Count each of the rectangles below to determine how many squares make up the area.



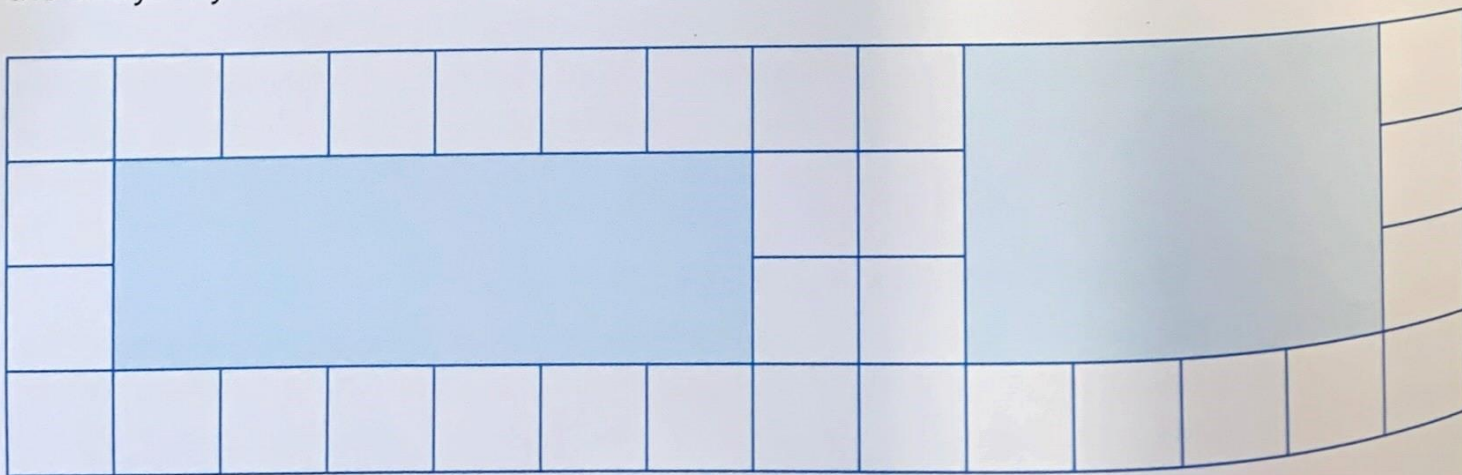
Using your knowledge of arrays, what would be another way that you could determine the area of each without having to count every square?

Explore

Find the rectangle shown above that has a base of 6cm and a height of 3cm. How could you use the base and height measurements to determine the area in cm^2 ? Try your idea with the other rectangles shown (including the square as it is a special type of rectangle). What do you notice? How can you write your idea as a formula using the variables Area (A), Base (B) and Height (H)?

Evaluate

Is there a way that you could use your formula if you didn't know how many squares were in the array? Try it on the following rectangles.



Explain

Explain how you would use your formula and your ideas so far to calculate the area of the following rectangles:

1. Your desk in cm^2
2. Your classroom floor in m^2
3. A rectangle that was twice as long as it was high. The height was 5cm.
4. A square that was 6cm high
5. A square that was half a metre long – how many square metres does it take up altogether?

Extend

To measure area, we use square centimetres (cm^2), square metres (m^2) and hectares. These units increase in size at the same rate.

Use this idea to work out:

1. How many square centimetres fit in one square metre?
2. How many square metres fit in one hectare?
3. What would a hectare look like if it was a square?

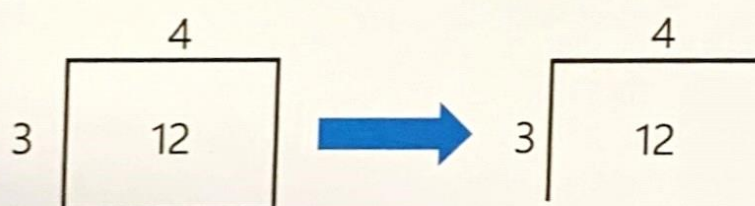
13. Division with arrays

MULTIPLICATIVE THINKING

*Multiplicative thinking underpins division, and forms a very strong link to fractions, decimals and percentage. In this activity, you will link your **Array Strategy** to division.*

Previously, we examined arrays made with 12 and with 36 squares. Division can be thought of as making arrays when you know the total amount of squares and the number of rows, and you have to find the missing number of columns (or vice versa).

Using this thinking, an array with 3 rows of 4 to make 12 can look like the first picture below. It also looks similar to the symbolic representation for short division shown in the second picture.



Experiment

In this activity, we will apply thinking about using the **Array Strategy** with grids of squares to dividing when the divisor is not a factor of the dividend. That means you will need to think about what to do with the left-over squares that do not fit. Use grid paper to draw a rectangle that has a total area of 12 squares, but with exactly 5 columns (5 squares are along the top of the rectangle). We can also write this as $12 \div 5$.

Explore

While you could leave the left-over squares as remainders, try instead to cut them into equal parts and spread the parts out between the columns. What have you tried? What has worked or not worked? You should be able to find at least two answers that use fractions of the square.

What do you notice about the fraction you now have and the numbers in the original question? How do you think fractions and division might be connected?

Try thinking about dividing \$12 between 5 people. How much money would each person get? Write it as a decimal number.

Evaluate

Fraction answer: If you have ended up with a mixed number (whole number and a fraction) as your answer, try turning it into an improper fraction.

Explain

Share all your answers as a group. Try to find connections between each answer. What patterns can you find?

Extend

Try to draw an array made of 10, with sides of only 3. What happens? What do you notice?

14. Division, fractions and decimals

 **MULTIPLICATIVE THINKING**  **PROPORTIONAL REASONING**

Work with a partner to look at the worked example, then apply your understanding of the connections between division, fractions and decimals to the following problem sets. Try to work out each answer mentally or by drawing first, then use a calculator to check. Division, fractions, decimals and percentages are all representations of the exact same amounts.

Strategies:

Array Strategy gives you a fraction answer.

Of A Dollar Strategy gives you a decimal answer.

Set A:

Give each answer as a remainder, fraction and decimal

1. $6 \div 4$
2. $6 \div 5$
3. $9 \div 4$
4. $9 \div 5$
5. $29 \div 2$
6. $29 \div 3$
7. $29 \div 5$

Set B:

There are times when the way you express a remainder matters. For each of the situations below decide which remainder is most appropriate and explain why.

8. Making teams of people from a class
9. Sharing cups of flour between bowls
10. Allocating money to budgets
11. Sharing cupcakes between friends
12. Sharing wrapped lollies between friends
13. Allocating time to different activities when there are not whole hours involved
14. Sharing a total land area among potential allotments
15. Sharing prize money between a team

Extend

Sometimes, people think that small numbers cannot be divided by larger numbers. This isn't true because we can split the small number into fractions or decimals. Try drawing an array to show a total area of 2 squares, with a length of 3 squares. This is the same as $2 \div 3$. You can think of it as 2 cakes shared between 3 people, or \$2 shared between 3 people if that is helpful. Draw the diagram and explain your thinking. Discuss what you have done with a friend.

 **MULTIPLICATIVE THINKING**
 **PROPORTIONAL REASONING**

$17 \div 4$ can be shown in different ways. All the following methods represent the same amount.

$$\begin{array}{r} 4 \text{ r}1 \\ 4 \overline{) 17} \end{array}$$

$$\begin{array}{r} 4 \frac{1}{4} \\ 4 \overline{) 17} \end{array}$$

$$\begin{array}{r} 4.25 \\ 4 \overline{) 17.00} \end{array}$$

15. Operations with multiple steps

GENERALISING

Work with a partner to look at the worked example, then figure out which of the orders given below for operations is correct. The correct answer will obey all four rules in the worked example. You may use a calculator as needed.

Evaluate

Which **one** of the following statements gives the correct order for operations?

- A. Brackets, indices, multiplication, then division, then addition, then subtraction
- B. Brackets, addition and subtraction as they appear, then multiplication and division as they appear
- C. Brackets, indices, then work left to right with whatever operation appears
- D. Brackets, indices, multiplication and division as they appear, then addition and subtraction as they appear

GENERALISING

First rule:

$$4 \times 5 \times (9 + 3) = 240$$

Second rule:

$$3 + (2+3)^2 = 28$$

Second rule:

$$4 + 5 + 3 \times 6 = 27$$

$$4 + 5 + 15 \div 3 = 14$$

Third rule:

$$5 \times 6 \div 2 \times 3 = 45$$

Fourth rule:

$$6 - 3 + 4 - 5 = 2$$

Set A:

Calculate the solutions for the following equations. You may use a calculator, but you need to show your working as well.

- 1. $7 \times 9 + (3 + 7) =$
- 2. $4 + 3 - 2^2 \times 3 =$
- 3. $(5 - 3)^2 \times 5 + 9 =$
- 4. $12 \times (3 + 2) \div 10 =$
- 5. $6 \times 7 \times 2 \div 12 =$
- 6. $12 \div 2^2 \times 5 =$
- 7. $12 + 14 \div 2 =$
- 8. $19 - 5 \times (7 - 4) =$
- 9. $7 - 4 + 7 - 3 - 1 =$
- 10. $12 + 4 \times 5 \div 2 - 11 =$

Set B:

Write equations for the following situations. Make sure that you use correct order of operations. Solve each equation and show your working.

- 11. Jacki had three prepaid SIM cards worth \$25 each. She spent \$67 on phone calls. How much does she have left to spend?
- 12. Michael had 12 chocolates which he divided into 3 bags. Then he added another 5 chocolates to one of the bags. How many chocolates were in that bag?
- 13. The price of a shirt was marked at \$5 off from the original price of \$20. Janessa bought 6 shirts. How much did she spend?

Set C:

Write a situation for each equation. You do not need to solve them.

- 14. $(60 + 3) \div 7$
- 15. $35 \div 5 - 4$
- 16. $7 + (8 \times 3) - 20$

16. Approximation using leading digits

GENERALISING

*Being able to approximate is incredibly useful. It allows you to quickly determine whether an answer is reasonable, for example, when considering a quote for work from a tradesperson. One of the easiest ways to approximate is by rounding all numbers to the leading digit, then performing the operations. Use the worked example below to figure out **Leading Digit Approximation** works. This strategy links with previous learning from lessons 2 and 4.*

Set A: Whole numbers

1. 345×920
2. $354 + 290$
3. $354 - 902$
4. $345 \div 920$

Set B: Decimal numbers

5. 3.45×92.0
6. $35.4 + 2.90$
7. $354 - 9.02$
8. $34.5 \div 9.20$

Set C: Multi-step equations

9. $35 \times 980 + 0.87$
10. $354 + 290 \times 0.87$
11. $35^2 - 902 + 78$
12. $345 \div (920 \div 3.29)$

Worked examples

$$\begin{aligned} 23\,432 \times 558 \\ \cong 20\,000 \times 600 \\ \cong 12\,000\,000 \end{aligned}$$

$$\begin{aligned} 2.95 \times 0.845 \\ \cong 3 \times 0.8 \\ \cong 2.4 \end{aligned}$$

$$\begin{aligned} 967 \div 18 \\ \cong 1000 \div 20 \\ \cong 50 \end{aligned}$$

$$\begin{aligned} 33.4 \times 1.9 - 4.45 \\ \cong 30 \times 2 - 4 \\ \cong 56 \end{aligned}$$

Set D: Application – approximation and reasonableness

Sometimes Leading Digit Approximation method is not appropriate for the situation as it results in an unreasonable answer. Complete the following approximations and comment on the **reasonableness** of the answers.

13. The Adelaide oval has a capacity of 53 500 people. It is approximately 95% full. How many people are there?
14. The mass of a ute is 1 837kg. The mass of its load is 930kg. How much does it weigh?
15. In 2019, 15 229 Australians were diagnosed with melanoma. Chance of surviving at least five years is 91%. Approximately how many will be still alive in 2024?
16. According to the Centre for Disease Control and Prevention, 36 000 Americans are killed by guns each year. What is the average number of people killed by guns per day?

Evaluate

What did you like about this strategy? What didn't you like?

When do you think this strategy would be useful?

When would you use something else instead?

17. Percentage of an amount

 MULTIPLICATIVE THINKING  PROPORTIONAL REASONING

Percentages are one of the most commonly used mathematical concepts outside of a school context. In this activity, you will work with a friend to develop your understanding of percentages, then apply them to answer the questions below.

Stimulus questions: finding a percentage of an amount

10% of 4 = 0.4

- How do we write 10% as a decimal number?
- What operation does "of" usually mean? For example, 2 packets of 10 lollies.
- Use your calculator to determine how to find 10% of 4. The answer should be 0.4.

75% of 20 = 15

- How do we write 75% as a decimal number?
- Use your calculator to determine how to find 75% of 20.
- Alternatively, you could think about 75% as being the same as $\frac{3}{4}$ and work from there.

Stimulus questions: finding what percentage of an amount another amount makes up
15 out of 20 is 75%.

- How do we write 15 out of 20 as a fraction?
- What operation does "out of" usually mean? For example, 2 out of the 10 lollies were green.
- How do we write that fraction as a decimal, then a percent? Use your calculator to determine how to use the numbers 15 and 20 to get 75% or 0.75.

10 out of 40 is 25%.

- How do we write 10 out of 40 as a fraction?
- What operation does "out of" usually mean?
- How do we write that fraction as a decimal, then a percent? Use your calculator to determine how to use the numbers 10 and 40 to get 25% or 0.25.

Set A: Mixed practice and application

1. Find 70% of 40
2. What percentage is 45 out of 50?
3. Jen scored 80% on her exam. There were 20 questions. How many did she get correct?
4. Jen scored 12 out of 20 on her exam. What was her percentage correct? What was her percentage incorrect?
5. A shirt was marked down from \$30 to \$24. What percentage was the discount?
6. The interest rate for a bank loan was 4% per year. The loan was for \$200 000. How much interest will be charged for one year?
7. A couple paid \$15 000 interest on their loan of \$300 000 last year. What was the interest rate as a percentage?
8. The superannuation rate is 9.5%. An employee earns \$45 000 per year plus the 9.5% superannuation. How much superannuation is paid?



18. Percentage off

 PARTITIONING  PROPORTIONAL REASONING

Most people use quite complicated procedures when finding how much money they will pay for a discounted amount. This activity will teach you a much simpler way of thinking about discounts.

Stimulus questions: finding a percentage *off* an amount

25% off \$20 = \$15

- We could try finding 25% of \$20 (which is \$5), then taking that amount away from the \$20.
- An alternative is to use our **Partition It** to work out the percentage that would be left. If we took 25% away from 100%, *what percent is left?*
- Try finding that percent of \$20 using your calculator or a mental strategy.
- What do you notice?

20% off \$30 = \$24

- We could try finding 20% of \$30, then take that amount away from the \$30.
- An alternative is use our **Partition It** to work out what would be left. If we took 20% away from 100%, *what percent is left?*
- Try finding that percent of \$30 using your calculator or a mental strategy.
- What do you notice?

Set A: Quick practice

1. Find 70% off \$40
2. Find 30% off \$60
3. What percentage discount was given if a shirt now costs \$45, reduced from \$50?
4. What percentage discount was given if a shirt is \$10 off the initial price of \$80?

Set B: Complex Application

The following questions have a mix of percentage *of* and percentage *off*, with multiple steps and some that require you to work backwards. Diagrams will help. Work with a friend if you find it helpful. Use the **Partition It** strategy if it helps.

5. Hannah earned 20% of the total tips and Adam earned 30% of the tips. Adam was paid \$45 for his 30%. How much was Hannah paid? What was the total amount tipped over the course of the shift?
6. Laura bought 3 shirts for 50% off and paid a total of \$37.50. What was the original price of each shirt?
7. 25% of the children in the class were home with chicken pox. Another 15% were at choir practice. That left Elise and 17 others. How many people were in the class normally?
8. A jar contained 50 jellybeans. 10% of the jellybeans were strawberry flavoured. Liam only liked the black jellybeans, so he ate all 8 of them. How many jellybeans were **not** black or strawberry? What percentage is that?

19. Introducing ratios

 MULTIPLICATIVE THINKING  PROPORTIONAL REASONING

Ratios are one of the most commonly used mathematical tools in the real world. They are used to represent the size of proportions in relation to each other, e.g. proportion of ingredients in a recipe, the length of fabric needed for making a curtain compared to the length of the window.

Explore

Use the worked example to identify how ratios are *similar* to and *different* from fractions.

- List the similarities and differences that you can find.
- How do you think ratios work?

What would change if you needed to cook 4 cups of rice?

- How much water would you need to use?
- How do you know? What did you do to work it out?

What would change if you needed to cook 1 cup of rice?

- How much water would you need to use?
- How do you know? What did you do to work it out?

Worked example

To cook most types of rice we use a ratio to determine how much water to add. For every **2** cups of rice, we add **3** cups of water. We use **5** cups of ingredients in total.

The **ratio** is 2:3

The **fraction** of rice is $\frac{2}{5}$

The **fraction** of water is $\frac{3}{5}$

Apply

Ratios focus on the parts **in relation to each other**. **Fractions** focus on the parts **in relation to the size of the whole**. Try using what you have worked out about ratios to determine how to express the following ratios.

1. When determining how much fabric to buy for curtains, we need to allow extra fabric for the folding or curving. We need 3 times the window length of fabric.
 - a. Write this as a ratio.
 - b. How much fabric would we need if the window was 4m long?
 - c. If we bought 18m of curtain fabric, how long would the window be?
2. When cooking a meal, we need 1 cup of rice for every 2 adults.
 - a. Write this as a ratio.
 - b. How much rice would we need for 6 adults?
 - c. If we cooked 2 cups of rice, how many adults could we feed?

Extend

3. In a class of students, 2 children played cricket for every 3 children who played soccer.
 - a. Write this as a ratio.
 - b. If there were 25 children in the class in total, how many played soccer and how many played cricket? How did you work it out?
4. A bag of lollies had a ratio of red: yellow: blue of 4:3:2
 - a. If there were 9 yellow lollies, how many red and blue were there? Explain why.
 - b. If there were 36 lollies altogether, how many would there be of each colour?

20. Ratios and recipes

🔗 MULTIPLICATIVE THINKING 🔗 PROPORTIONAL REASONING

We only write ratios using whole numbers. We wouldn't write $1:1\frac{1}{2}$ as a ratio of rice to water (1 cup of rice needs $1\frac{1}{2}$ cups of water). Instead, we would double both numbers and write it as 2:3 (2 cups of rice need 3 cups of water). When writing ratios, we use the same measuring unit for each part (e.g. using cups: cups, not cups: litres).

Experiment

Examine the recipe for playdough provided. The ratio of flour, salt and oil is easy to calculate as all the ingredients are measured in cups. The oil is more difficult as it is measured in millilitres. You will need to know that 1 cup holds 250mL.

Work out a ratio for the ingredients using the same base measurement.

Adjust your ratio so that the oil is 1. How do your other ingredients change?

Recipe for playdough

- 2 cups plain flour
- 1 cup salt
- 25mL oil
- 1 cup cold water
- Food colouring if desired

Mix flour and salt.
Add wet ingredients.
Knead until combined.

Explore

Stimulus questions

- How many millilitres of water is used in the original recipe?
- How much more water is used than oil?
- How much more flour is used than oil?
- How much more salt is used than oil?
- If we used only 1mL of oil, how much would we use of the other ingredients?

What have you tried? What has worked or not worked?

Evaluate

Does your ratio use only whole numbers?

Does your ratio use the same system for measurement?

How reasonable is your answer?

Change anything that you need to.

Explain

Share your answers as a group. You may have used different methods to work out the answers. Share how you did it and try to find similarities and differences in your strategies.

Extend

A kindergarten wanted to make roughly 1 cup of playdough for each child. There are 18 children in the class. How much of each ingredient should they use?

22. Dividing quantities into ratios

 MULTIPLICATIVE THINKING  PROPORTIONAL REASONING

Sometimes, we have to use the total number of items and a ratio to work out how many items there will be of each type. This is called dividing quantities into ratios.

Experiment

In our introductory lesson on ratios, the last question asked you to work out how many of each colour of lolly there would be when you knew how many lollies were in the bag and the ratio of colours. This is called dividing quantities into ratios. In this lesson we will learn a strategy for how to calculate the number of each item, using a **Relationship Table**.

The question you answered previously was:

A bag of lollies had a ratio of red: yellow: blue of 4:3:2. If there were 36 lollies altogether, how many would there be of each colour? The answer was 16 red, 12 yellow and 8 blue.

Red	Yellow	Blue	Total
4	3	2	9
			36

How did we change the 9 into a 36?
Apply that same change to all other numbers to see what happens.

Explore

Stimulus questions

- If we add up the numbers in the original ratio, what is the total?
- How is that number related to 36?
- How could we have used that number to change our 4 red into 16 red?
- Does that same process work for the other numbers?

Try to summarise a process for calculating this number, then using this number to divide the quantity of 36 into the ratios.

Extend

In the 2016 Australian census, 52% of respondents identified as Christian, 30% stated that they had no religion, 10% of people chose not to answer the question, and the remaining 8% identified mostly with Islam, Buddhism, Hinduism Sikhism and Judaism.

1. Write this as a ratio of Christians: No religion: No answer provided: Other religions, using everything you have learned about ratios so far.
2. Currently the Australian population is around 25 million. Use this to predict how many Australians there would be for each of the religions mentioned.

Evaluate

Try using your method to solve these questions and evaluate how well it works.

- A bag had 30 lollies in the ratio of 2 red to 3 blue. How many were there of each?
- A bag had 24 lollies in the ratio of 2 red to 1 blue. How many were there of each?

How well did your method work?

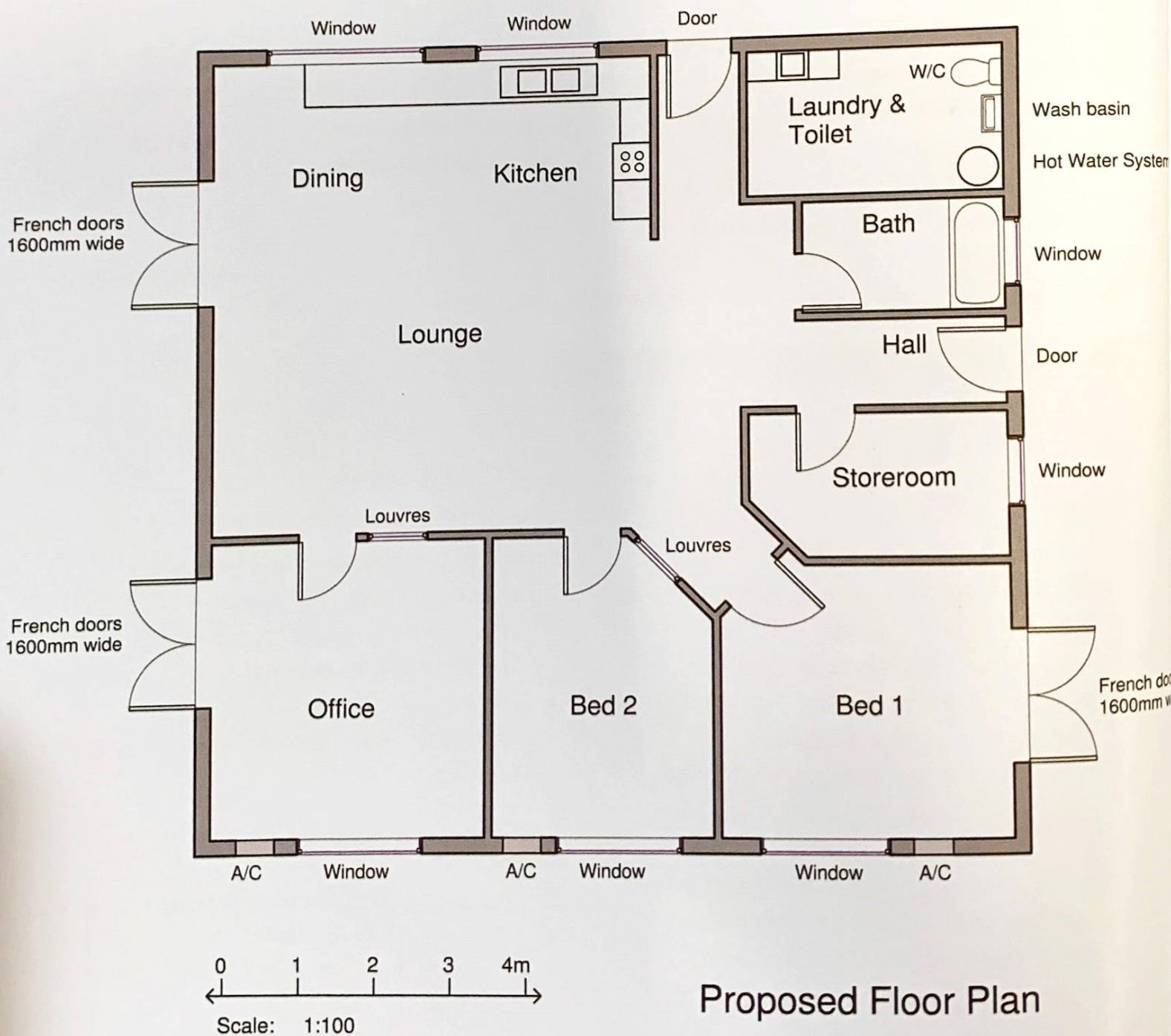
Explain

Write a simple process for dividing a quantity into a ratio. Try to explain your process in as few steps as possible, making sure that it always works.

23. Ratios as scales

 MULTIPLICATIVE THINKING  PROPORTIONAL REASONING

One of the most common applications for ratios is for scales on maps and plans. The diagram below shows a proposed floor plan for a house. The scale is given as a ratio at the bottom. In this case, the scale means that 1cm on the plan is the same as 100cm (or 1m) in real life.



Set A: Practice and application

1. Give the dimensions for the following rooms: office, bath, laundry.
2. Using the same scale, draw your desk. Commonly house plans are drawn using a scale of either 1:100 (above) or 1:50. Use a scale of 1:50 to draw your desk. Compare the drawings.
3. A city road map was drawn using the scale 1:50 000. Explain what this means. If you drove a road that was 4km long, how far would that be on your map?
4. Walking maps tend to use a scale of 1: 25 000. How many cm would be required to show a walking distance of 1km? Why is this important? Why would this scale be used?

24. Rates are used for different units of measure

 **MULTIPLICATIVE THINKING**  **PROPORTIONAL REASONING**

Rates are very similar to ratios in that they express the relationship of parts. However, whereas ratios compare the same unit of measurement, rates compare different units of measurement. For example, a student typed 80 words in 2 minutes. We usually express rates as "per one" of a measuring unit. This means that typing speed would be described in "words per minute" rather than words per two minutes. 80 words typed in 2 minutes would be expressed as 40 words per minute (40wpm or 40 w/m).

Experiment

Examine the rates below. Work out how to express them properly. Use the commonly used rates information to help you work them out.

1. A car travelled 120km in 2 hours. Write the car's speed in km/h.
2. A car travelled 25km in 30 minutes. Write the car's speed in km/h.
3. A plumber earned \$15 in 15 minutes. Write the plumber's pay rate per hour.
4. A lamp used 6kw of power in 3 hours. Write the lamp's energy consumption rate in kw/h.
5. A mobile phone user was charged 30c (plus a standard connection fee) for a 15 minute call to the USA. Write the phone charge as a rate per minute.

Commonly used rates

- Car speed: km per hour (kph or km/h)
- Pay rates, penalty rates and charge-out rates: \$ per hour
- Currency exchange rates: amount per \$1AUD
- Motor speed: revolutions per minute (RPM)
- Interest rates: interest per \$1
- Energy consumption rates: kilowatts per hour (kw/h)
- Phone charges: \$ per minute

What did you do to write each of the situations above as a rate "per one"? Try to explain the process in a way that can be easily simplified.

Explore and Evaluate

The **Relationship Tables** below might help you to work out a method for expressing unit rates correctly. Work through them until you think you have a method that works consistently. The questions below use the same numbers as above so that it makes more sense to you.

km	h
120	2
60	1

km	h
25 km	$\frac{1}{2}$ hour
	1

\$	h
15	$\frac{1}{4}$ hour
	1

kw	h
6	3
	1

Explain

How did you use multiplication or division in your Relationship Tables to write the rates properly? What processes did you use? Why did you have to start with the "per one" unit? Explain your system as simply as possible.

25. Calculating with rates: speed, rainfall, pay rates

 MULTIPLICATIVE THINKING  PROPORTIONAL REASONING

Once we know a unit rate, we can use it to easily calculate. In this task, you will first calculate a unit rate using your method from last lesson, then apply it to real world situations.

Explore and Apply

Use **Relationship Tables** to answer the following questions.

1. An apprentice carpenter earned \$80 in 5 hours. What was her pay rate? What would she have earned in a 38 hour week?

\$	h
80	5
	1

So

\$	h
	1
	38

2. A casual teacher earned \$456 for a 6 hour day. What was her pay rate? What would she have earned in a week of work (30 hours)?

\$	h
456	6
	1

So

\$	h
	1

3. In 2019, Townsville received a record amount of rainfall over the ten days leading up to 6th February. According to the Bureau of Meteorology, 1257mm fell in 10 days¹. What was the average rate of rainfall per hour for the whole ten days (240 hours)? Round the rate off to the nearest whole millimetre.

mm	h
1257	240
	1

4. In 1997, Andy Green set a new land-speed record when driving a twin turbofan jet-powered car. He averaged 1227.985 km/h over the course². The course was 1 mile, which is 1.61km long. How long would it have taken him to drive the 1.61km? Try to answer in hours (this will be very small), then convert it into minutes and seconds using similar thinking. You should end up with a final answer of less than 5 seconds.

km	h
1228	1
1.6	

Or...

km	min
1228	60
1.6	

Or...

km	sec
1228	3600
1.6	

¹ <http://www.bom.gov.au/climate/current/statements/scs69.pdf>

² <https://www.fia.com/>

26. Converting between rates 1

 MULTIPLICATIVE THINKING  PROPORTIONAL REASONING

*In the real world, we often have to convert between different rates to make comparisons. Sometimes, we only have to convert one of the parts, so we perform one operation. This activity looks at converting one part only. Use **Relationship Tables** to help you.*

Experiment: Converting with one part

Speed in the USA is expressed in miles per hour (mph or mi/h) whereas, in Australia, we use kilometres per hour (kph or km/h). In this case, we need to convert the distance, but not the time as both are expressed "per hour". Look at the table on the left below and use it to work out how to change 60 miles per hour into km per hour in the table on the right.

distance	time
1 mile	1h
= 1.6 km	1h

unit rate

distance	time
60 miles	1h
km	1h

Explain

Complete these sentences:

- To convert from 1 mile to km, we have to multiply by 1.6, so to convert from 60mph to kph we have to...
- To convert from 1km to miles we have to... so to convert from kph to mph we have to...

Apply

Apply your process to convert the following rates. All of the questions require you to only convert the distance. To help you out here are some imperial to metric conversions. Each have been rounded to a single decimal place.

$$1 \text{ mile} = 1.6\text{km} \quad 1 \text{ yard} = 0.9\text{m} \quad 1 \text{ inch} = 2.5\text{cm}$$

Animal	Top speed in imperial units ³	What do I do?	Speed in metric units
Sea star	0.6 miles per hour		km/h
Snail	30 inches per min		cm/min
Sloth	41 yards per day		m/day
Manatee	5 miles per hour		km/h
Slug	1.9 miles per hour		km/h

- What is the same about your process each time? What would be different if we were converting from metric units to imperial units?
- Convert 25cm/min to inches/min.
- Convert 12m/day to yards/day.

³ <https://frontier.ac.uk/blog/2016/12/18/10-slowest-animals>

27. Converting between rates 2

 **MULTIPLICATIVE THINKING**  **PROPORTIONAL REASONING**

In the previous task, we converted one of the parts in a rate to a different unit (e.g. miles per hour to kilometres per hour). In this task, we will convert both of the units (e.g. km/h to m/sec). That requires performing two operations rather than one. Use **Relationship Tables** to help you.

Experiment

The following tables show what happens when we convert 1km/h into m/sec. Use the tables to work out what operations were performed to change the rate.

Step 1: Distance

distance	time
1 km	1h
= 1000 m	in one hour

old rate

new rate

Step 2: Time

distance	time
1000 m	in 3600 sec
	in 1 second

Explore

1. What operation did we perform to turn 1km into 1000m? What did we multiply by?
2. What operation did we perform to turn 3600 sec into 1 sec? What did we divide by?
3. Check your answer. You should have found that 1km/h is the same as 0.27 m/sec.

Evaluate

Check that your system is working by converting 60km/h into m/sec. You should get 16.6m/sec. If you are confident that it's working, try converting the following speeds from km/h into m/sec. Questions 6 and 7 convert from m/sec into km/h. Draw tables, if it will help, or just use operations if you find that easier. Round off your answers to a single decimal point.

4. 50km/h to m/sec
5. 80km/h to m/sec

6. 20m/sec to km/h
7. 10m/sec to km/h

Apply and Extend

The following questions are more complex as they require multiple steps. Remember to check your answers as you go.

8. I walked 12km in 2 hours. What is my speed in km/h? What is my speed in metres per minute? How far will I have walked after only 15 minutes?
9. I run 100m in 15 seconds. What is my speed in m/sec? What is my speed in km/hour?

28. Using rates to make comparisons

 MULTIPLICATIVE THINKING  PROPORTIONAL REASONING

*As rates are usually expressed "per unit", they are a great way of making comparisons. In this activity, you will apply what you have learned about rates to compare them. Use **Relationship Tables** to help you.*

Apply

Work out the rate for the price/kg for each of the following items and order them from the cheapest to the most expensive. Show your working.

1. A shopping centre sold salami in the following sizes:
 - A. 200g knob of salami for \$6.20
 - B. 80g packet of pre-sliced salami for \$3.25
 - C. Pre-sliced salami from the deli for \$32 per 100g
2. Oranges were sold in the following units:
 - A. A 3kg bag of oranges for \$5.50
 - B. Unpacked oranges for 60c each (oranges weigh on average 180g)
3. Tinned tomatoes came in the following sizes.
 - A. 600g diced cherry tomatoes for \$3.15
 - B. 400g diced cherry tomatoes for \$2.10

Extend

When discounts are applied to items, it changes their price rate. How would your answers to the questions above change if a 10% discount was applied to the most expensive item in each group? Show your working.